

S. N.	Details of Enclosure	Status of Submission of information/ documents	Remarks
	<i>[Name of Documentary evidence in support of compliance]</i>		
(vii)	Indian Petroleum Act 1934 or latest and Rules. — <i>[Name of Documentary evidence in support of compliance]</i>	Yes/No	
(viii)	License under the contract Labour (Regulation & Abolition) Act 1970 or latest and Rules. — <i>[Name of Documentary evidence in support of compliance]</i>	Yes/No	
(ix)	Indian Electricity Rule 2003 and amendments if any, from time to time. — <i>[Name of Documentary evidence in support of compliance]</i>	Yes/No	
(x)	The Environment (Protection) Act 1986 or latest and Rules. —	Yes/No	

S. N.	Details of Enclosure	Status of Submission of information/ documents	Remarks
	[Name of Documentary evidence in support of compliance]		
(xi)	Child Labour (Prohibition & Regulation) Act 1986 or latest. — [Name of Documentary evidence in support of compliance]	Yes/No	
(xii)	National Building Code of India 2005 or latest (NBC 2005). — [Name of Documentary evidence in support of compliance]	Yes/No	
(xiii)	Indian standards for construction of Low/ Medium/ High/ Extra High Voltage Transmission Line — [Name of Documentary evidence in support of compliance]	Yes/No	
(iv)	Any other statutory requirement(s) [please specify] — [Name of Documentary evidence in support of compliance]	Yes/No	

S. N.	Details of Enclosure	Status of Submission of information/ documents	Remarks
13.	Annexure – 10B (SP) Details of Insurance Policies along with documentary evidences taken by the Contractor for the insurance coverage against accident for all employees as below:		
(i)	Under Workmen Compensation Act 1923 or latest and Rules. — <i>[Name of Documentary evidence in support of insurance taken]</i>	Yes/No	
(ii)	Public Insurance Liabilities Act 1991 or latest — <i>[Name of Documentary evidence in support of insurance taken]</i>	Yes/No	
(iii)	Any Other Insurance Policies — <i>[Name of Documentary evidence in support of insurance taken]</i>	Yes/No	

18. FORM OF JOINT DEED OF UNDERTAKING BY THE SUB-CONTRACTOR ALONGWITH THE CONTRACTOR

THIS DEED OF UNDERTAKING executed this day of Two Thousand and by M/s., a Company incorporated under the laws of and having its Registered Office at (hereinafter called the “**Sub-Contractor**” which expression shall include its successors, executors and permitted assigns), and M/s., a Company incorporated under the laws of having its Registered Office at (hereinafter called the “**Bidder**”/”**Contractor**” which expression shall include its successors, executors and permitted assigns) in favour of **XXXX (Name of the Employer)** a Company incorporated under the Companies Act of 1956/2013 (with amendment from time to time) having its registered office at **XXXX (Registered Address of the Employer)** (hereinafter called the “**Employer**” which expression shall include its successors, executors and permitted assigns)

WHEREAS the “**Employer**” invited Bid as per its Specification No.....for Development of Distribution Infrastructure & System Strengthening works at Zone of MSEDCL, Maharashtra under Power Distribution Enhancement Program funded by Asian Development Bank (ADB) **including installation of Sub-stations, lines, bays, DTs and providing service connections etc.**

AND WHEREAS Clause No., Section, of, Part... forming part of the Bid Documents inter-alia stipulates that the Bidder and/or Sub-Contractor must fulfill the Qualifying Requirements and be jointly and severally bound and responsible for the quality and timely execution of System Strengthening works funded by ADB in the event the Bid submitted by the Bidder is accepted by the Employer resulting in a Contract.

AND WHEREAS the Bidder has submitted its Bid to the Employer vide Proposal No. dated based on tie-up with the **Sub-Contractor** for execution of aforesaid System Strengthening works funded by ADB.

NOW THEREFORE THIS UNDERTAKING WITNESSETH as under:

- 1.0 In consideration of the award of Contract by the Employer to the Bidder (hereinafter referred to as the “**Contract**”) we, the **Sub-Contractor** and the **Contractor** do hereby declare that we shall be jointly and severally bound unto the **XXXX (Name of the Employer)**, for **execution of** System Strengthening works funded by ADB in accordance with the Contract Specifications.

- 2.0 Without in any way affecting the generality and total responsibility in terms of this Deed of Undertaking, the **Sub-Contractor** hereby agrees to depute their representatives from time to time to the Employer's Project site as mutually considered necessary by the Employer, Contractor and the **Sub-Contractor** to ensure proper quality, manufacture, testing and supply on FOR destination delivery at site basis and successful performance of System Strengthening works funded by ADB in accordance with Contract Specifications. Further, if the Employer suffers any loss or damage on account of non-performance of the material fully meeting the performance guaranteed as per Bid Specification in terms of the contract. We the **Sub-Contractor** and the Contractor jointly and severally undertake to pay such loss or damages to the Employer on its demand without any demur.
- 3.0 This Deed of Undertaking shall be construed and interpreted in accordance with the laws of India and the Courts in xxxxx (Headquarter of Employer) shall have exclusive jurisdiction in all matters arising under the Undertaking.
- 4.0 We, the Bidder/Contractor and **Sub-Contractor** agree that this Undertaking shall be irrevocable and shall form an integral part of the Contract and further agree that this Undertaking shall continue to be enforceable till the Employer discharges it. It shall become operative from the effective date of Contract.

IN WITNESS WHEREOF the **Sub-Contractor** and/or the Contractor have through their Authorised Representatives executed these presents and affixed Common seals of their respective Companies, on the day, month and year first above mentioned.

WITNESS

(For **Sub-Contractor**)

Signature

(Signature of the authorized representative)

Name

Office Address

Name

Common Seal of Company
.....

WITNESS

(For Bidder)

Signature
 Name (Signature of the authorized representative)
 Office Address Name
 Common Seal of Company

Note:

1. For the purpose of executing the Deed of Joint Undertaking, the non-judicial stamp papers of appropriate value shall be purchased in the name of executant(s).
2. The Undertaking shall be signed on all the pages by the authorised representatives of each of the partners and should invariably be witnessed.
3. This Deed of Joint Undertaking duly attested by Notary Public of the place(s) of the respective executant(s), shall be submitted along with the bid.
4. In case the bid is submitted by a Joint Venture (JV) of two or more firms as partners, then the Joint deed of undertaking shall be modified accordingly.

19. FORM OF CERTIFICATE OF FINANCIAL PARAMETERS FOR QR**(as per clause ref. no. 2 of Section 2 of Part 1))****(Rupees in Lakhs)**

S. No.	Financial parameters	2025-26	2024-25	2023-24	2022-23	2021-22
1.	Net Worth					
a)	Paid up Capital					
b)	Free Reserves and Surplus*					
c)	Misc. expenses to the extent not written off					
	Net Worth (a+b-c)					
2.	Annual Turnover **					
3.	Liquid Asset (Total Current Asset – Inventories)					

* Free Reserve and Surplus should be Exclusive of Revaluation Reserve, written back of Depreciation Provision and Amalgamation.

** Annual total Income/ turnover as incorporated in the Profit and Loss Account excluding non-recurring income, i.e., sale of fixed asset etc.

It is certified that all the figures are based on audited accounts read with auditors report and Notes to Accounts etc.

Date

Place

Certified By

(Chartered Accountants)

Membership No.

Seal

20. FORM OF ATTESTATION LETTER FROM THE BIDDER TO THE MSEDCL**Attestation Letter**

Date: OCB No.:

To: Maharashtra State Electricity Distribution Company Limited

Plot G-9 , “Prakashgad” , 6th Floor, Anant Kanekar Marg, Bandra (E), Mumbai – 51

We, the undersigned, acknowledge that any of the activities or production involving forced labor¹ or child labor² is a sufficient ground for bid rejection and may give MSEDCL the right to terminate the contract with us.

After duly considering the above policy,

(a) we attest and represent that (i) to the best of our knowledge, we are not using (and we have not used, and we will not use) works, goods and services, and related materials and products in the Project, where the production and supply of such products used forced labor or child labor; (ii) we adhere to labor legislation of the host country and require our subcontractors and suppliers for the Project to adhere to the same in the production and supply of works, goods and services, and related materials and products proposed to be used in the Project; and (iii) to the best of our knowledge, the works, goods and services, and related materials and products to be procured and/or supplied by us for the Project do not involve production or activities involving the use of forced labor or child labor;

(b) we attest and represent that our subcontractors, manufacturers and suppliers have confirmed to us separately that to their knowledge, after due enquiry and diligence, the offered works, goods and services, and related materials and products do not involve production or activities involving the use of forced labor or child labor;

¹ Forced labor means all work or services not voluntarily performed, that is, extracted from individuals under threat of force or penalty.

² Child labor means the employment of children whose age is below the host country's statutory minimum age of employment or employment of children in contravention of International Labor Organization Convention No. 138 “Minimum Age Convention” (www.ilo.org).

(a) we further confirm that should MSEDCL require an audit of our records related to the

procurement of works, goods and services, and related materials and products used in the Project, we shall promptly make available MSEDCL or your respective designated representatives all relevant documents and records to assist with the audit, and grant access, to the extent practicable, to the sites, facilities, plants, and equipment to an independent auditor retained by MSEDCL;

- (b) we confirm that if the contract is awarded to us, (i) we shall monitor the works, goods and services, and related materials and products provided by us on an ongoing basis, (ii) require our subcontractors, suppliers or manufacturers to immediately notify to us any incidents of forced labor or child labor, and if new risks or incidents of forced labor or child labor are identified, (iii) we commit to promptly inform you, if we receive information that the representation in (a) above is false and of any new risks or incidents of forced labor or child labor in the production or activities for the offered works, goods and services, and related materials and products used in the Project, and to take appropriate steps to remedy them;
- (c) we agree that a breach and misrepresentation of (a), (b), (c), or (d) above is a sufficient ground for bid rejection and may give the MSEDCL the right to terminate the contract with us, if our bid is accepted and the contract is awarded to us.

Sincerely,

Authorized Signature [In full and initials]: ---- Name and Title of
Signatory: _____

Address: _____

Social, Gender and Accessibility Compliance Requirements

ANNEX – A:

Contractor-Led Awareness Training (Community and Workforce)

The Contractor shall conduct structured community and workforce awareness trainings in accordance with ADB's gender, labour, and social safeguard requirements. The objective is to ensure that project beneficiaries and workers are informed about project objectives, grievance mechanisms, safety practices, and gender inclusivity measures.

1. Mandatory Training Requirements

THEME	TARGET GROUP	KEY CONTENT
PROJECT INFORMATION & BENEFITS	Farmers, PRI members, SHGs, CBOs	Project scope, feeder model, benefits, timelines, land use clarifications
LABOUR RIGHTS & SITE SAFETY	All site workers (incl. women)	PPE usage, wage parity, site safety, code of conduct, zero harassment policy
GENDER INCLUSION & PARTICIPATION	Women farmers, SHGs, female workers	Economic opportunities, O&M participation, grievance access
SOCIAL & ENVIRONMENTAL SAFEGUARDS	Local communities, labour groups	Dust/noise management, waste handling, cultural sensitivity
GRIEVANCE REDRESS MECHANISM (GRM)	All stakeholders	GRM procedure, helpline, verbal/written complaint options

Minimum Training Frequency:

- The Contractor shall provide minimum 2 trainings half yearly (Separately for each contract), ensuring at least 30% women participants.
- The Training materials must be in Marathi, using pictorial/visual IEC formats.

2. Contractor Obligations

- The Contractor shall submit a Training Implementation Plan within 15 days of contract signing.
- The Contractor shall conduct all trainings at worksites and community/village locations as per the approved plan.
- The Contractor shall maintain verifiable records including gender-disaggregated attendance, geo-tagged photos, and feedback.
- The Contractor shall include women farmers and SHG representatives in all community-level sessions.
- Only MSEDCL/PMU-approved IEC materials shall be used during trainings.
- The Contractor shall bear all training costs (venue, materials, logistics, reporting, etc.).

Gender-Responsive and Disability-Inclusive Substation Specifications

All substation control rooms and related infrastructure shall be designed and constructed to be gender-responsive and accessible to persons with disabilities (PwD) in compliance with the Rights of Persons with Disabilities Act, 2016, the National Building Code, and ADB Gender and Universal Access Policy.

Minimum Infrastructure Requirements

FACILITY ELEMENT	REQUIREMENT	STANDARD / SPECIFICATION
SEPARATE SANITATION UNITS	1 Male + 1 Female PwD Toilet	Western WC, grab rails, disposal bin, lighting, water supply
ACCESSIBLE PATHWAYS	Step-free, non-slip flooring	Ramp $\leq 1:12$ gradient, tactile tiles, handrails
SECURITY & SAFETY	Adequate lighting, CCTV at entry	Min. 20 lux illumination, no dark corners
SIGNAGE	Marathi + English + pictograms	Reflective, ≥ 1200 mm height
EMERGENCY RESPONSE	First-aid box, extinguisher, emergency contact board	Mandatory prior to commissioning

Price Schedule Format-I

Project : works at Nagpur Zone of MSEDCL, Maharashtra under Power Distribution Enhancement Program funded by Asian Development Bank (ADB)

Tender No. CE-SPD/ ADB/ 2026-27/ T-13

Zone - Nagpur

Schedule 1: Grand Summary

District - Nagpur

(Bidder to Carry Quoted Prices forward from Schedules 2 and 3)

Schedule Number	Description of Activity	Employer's Estimated Cost (Rs. Lakhs)
1	Civil Works	12930.58
2	Supply and Installation of Plant and Equipment:	
a.	Part 1: Cost of Material	29858.24
b.	Part 2: Cost of Transportation, Installation, Erection, Commissioning & Guarantee	9758.03
Grand Total (Excluding Taxes)		52546.85

The Taxes chargeable to Employer will be paid as per the rates given below		
1	GST@ ____ on Supply of equipment produced in Maharashtra For supply of equipment produced outside Maharashtra	9458.43
Grand Total & Taxes		62005.29

I/ we hereby offer to complete the above described Works contained in the Schedules 2 and 3 together, for the Price of: _____ % (_____ percent) above / below the total tender cost of the Employer's Estimate.

Signature of Bidder:

Name & Designation:

Company:

Notes :-

- 1) The bidder should quote his % rate on the total tender estimated cost. Separate rates should not be quoted against Civil, Material & Services Components.
- 2) The quoted rate and price must be exclusive of GST
- 3) Any deviation from the instructions contained in these notes shall make the offer of the bidder as non-responsive.

Price Schedule Format-II (CIVIL)

Project: Development of Distribution Infrastructure & System Strengthening works at Nagpur
Zone of MSEDCL, Maharashtra under Power Distribution Enhancement Program
funded by Asian Development Bank (ADB)

Circle : Nagpur Urban Circle & Nagpur
Zone: Nagpur

Tender No. CE-SPD/ ADB/ 2026-27/ T-13

Schedule 2: Civil Works and Other Services

Activity Number	Description of Activity	Unit	Quantity	Employer's Estimate	
			NZ Total	Unit Rate	Total Cost NZ
				(Rs. Lakhs)	(Rs. Lakhs)
(1)	(2)	(3)	(6)	(7)	(10) = (6) x (7)
0101	Civil works for 1 x 5 MVA S/S Conventional Type	Nos	5.00	102.58	512.92
0103	Civil works for 2 x 5 MVA S/S Conventional Type	Nos	3.00	111.39	334.16
0104C	Civil works for 2 x 10 MVA S/S Conventional Type	Nos	1.00	87.97	87.97
7004	Civil Works for New 33/11KV GIS-Substation 2 x 5 MVA	Nos	5.00	72.93	364.64
7005	Civil Works for New 33/11KV GIS-Substation 1 x 10MVA & NIFS System	Nos	1.00	72.93	72.93
7006	Civil Works for New 33/11KVGIS STN-- 33/11 KV 1 x 10 MVA (33 KV GIS & 11 KV AIS) & NIFS System	Nos	5.00	72.93	364.64
7007	Civil Works for New GIS 33/11 KV 2 x 10 MVA SStn & NIFS System	Nos	1.00	80.84	80.84
7008	Civil Works for New GIS 33/11 KV 2 x 10 MVA (33 KV GIS & 11 KV AIS)& NIFS System	Nos	2.00	80.84	161.68
0202	33/11KV Augmentation of Power TF From 5 MVA to 10 MVA including NIFPS	Nos	3.00	1.71	5.14
301	Foundation for 1 x 5 MVA Additional Power Transformer Without Gantry Structure	Nos	2.00	8.80	17.60
0302	1 x 10 MVA Power Transformer	Nos	4.00	8.80	35.21
0302B	Foundation for Additional Power Transformer 1 x 10MVA	Nos	2.00	8.80	17.60
NIFPS	Civil work for 10MVA NIFPS (Make -CTR) for Additional & Aug PTR	Nos	9.00	1.71	15.42
601A	Civil work for Foundation of 33 kV feeder bay without Gantry Structure	Nos	26.00	0.60	15.63
603A	Civil work for Founadtion11 KV Feeder bay without Gantry Structure	Nos	2.00	0.60	1.20
601B	Civil work for Foundation of 33 kV feeder bay with Gantry Structure & PT	Nos	8.00	0.69	5.49
1227	Civil work for 315 KVA on Pinth & RMU	Nos	12.00	0.34	4.10
6203R1	Civil works for foundation structure for New RMU (SF6 Type) with 2 Isolator + 2 Breaker	Nos	115.00	0.24	28.08
6203R2	Civil works for foundation structure for New RMU (SF6 Type) with 3 Isolator + 2 Breaker	Nos	42.00	0.24	10.26
HDD Bore	HDD Bore for 33KV/11KV UG cable	Nos	312.90	34.50	10795.05
	Total cost in Lakhs (Excluding Taxes)				12930.58

Price Schedule Format-III (MATERIAL & SERVICES)								
Project:	Development of Distribution Infrastructure & System Strengthening works at Nagpur Zone of MSEDCL, Maharashtra under Power Distribution Enhancement Program funded by Asian Development Bank (ADB)							
Tender No.	CE-SPD/ ADB/ 2025-26/ T-13					Zone:	Nagpur	
Schedule 3: Material and Services								
Activity Number	Description of Activity	Unit	NZ Total-Qnty	Tender Cost (Amount in . Lakhs)				
				Part 1: Cost of Material Unit Rate	Part 2: Cost of Services Unit Rate	Total Cost of Material	Total Cost of Services	Total Cost of Services NZ
(1)	(2)	(3)	(4)	(5)	(6)	(8)='(4) x '(5)	(9)='(4) x '(6)	(10) = (9) + (8)
0101	New 1 x 5 MVA S/S Conventional Type	No.s	5.00	84.69	20.496	423.47	102.48	525.95
0103	New 2 x 5 MVA S/S Conventional Type	No.s	3.00	160.78	38.908	482.33	116.72	599.06
0104C	New 2 x 10 MVA S/S Conventional Type	No.s	1.00	248.86	60.224	248.86	60.22	309.08
7004	GIS 33/11 AIS KV 2 x 5 MVA (for other cities)	No.s	5.00	233.56	56.522	1167.80	282.61	1450.41
7005	GIS STN--33/11 KV 1 x 10 MVA including NIFPS	No.s	1.00	276.48	66.909	276.48	66.91	343.39
7006	GIS STN--33/11 KV 1 x 10 MVA including NIFPS	No.s	5.00	166.40	40.270	832.02	201.35	1033.37
7007	GIS 33/11 KV 2 x 10 MVA Sub station	No.s	1.00	441.69	106.889	441.69	106.89	548.58
7008	GIS 33/11 KV 2 x 10 MVA (33 KV GIS & 11 KV AIS)including NIFPS	No.s	2.00	326.25	78.952	652.50	157.90	810.40
0202	33/11KV Augmentation of Power TF From 5 MVA to 10 MVA including NIFPS	No.s	3.00	102.20	24.731	306.59	74.19	380.78
301	1 x 5 MVA Additional Power Transformer Without Gantry Structure	No.s	2.00	76.13	18.425	152.27	36.85	189.12
0302	1 x 10 MVA Power Transformer	No.s	4.00	114.19	27.634	456.75	110.53	567.29
0302B	1 x 10 MVA Additional Power Transformer for 11KV indoor Panel including NIFPS	No.s	2.00	123.70	29.934	247.39	59.87	307.26
0601A	33 kV feeder bay without Gantry Structure	No.s	26.00	9.23	2.233	239.89	58.05	297.94
0601B	33 kV feeder bay with Gantry Structure & PT	No.s	8.00	11.87	2.872	94.93	22.97	117.91
0601D	11 kV Feeder bay with take off structure & bus bar	No.s	3.00	8.31	2.010	24.92	6.03	30.95
603A	11 KV Feeder bay without Gantry structure	No.s	2.00	6.76	1.636	13.52	3.27	16.80
0702	Suspension type with 100 Sq. mm AAAC conductor on 152 X 152 mm 13 mtr RSJ	KM	124.60	9.70	3.39	1208.77	422.22	1630.99
703	33 KV S/C Line Pin type with100 mm2 AAAC lines on 152 X 152 mm 11 mtr. RSJ	Km	6.00	8.78	3.07	52.69	18.40	71.10
0706	Express/Highway Crossing with 100 sqmm conductor on 152 X 152 SJ 13 mtr Poles for One span of 30 mtr.	KM	2.00	1.93	0.67	3.85	1.35	5.20
0708	33kv, 3 core X 300 sqmm XLPE Underground Cable	KM	428.65	25.25	8.82	10825.17	3781.23	14606.40
0709	33kv, 3 core X 300 sqmm XLPE Underground Cable for Railway line crossing /NH Crossing	Nos	18.00	8.74	3.05	157.31	54.95	212.25
0709-R	33kv, 3 core X 300 sqmm XLPE Underground Cable for HIGHWAY crossing for 60 mtr span with isolator on DP structure with both side isolator./ Highway crossing	Nos	14.00	8.74	3.05	122.35	42.74	165.09
0711	Douple pole structure(cut point) of- 33 kv line using13 m long RSJ pole (152x152)	Nos	194.00	0.98	0.34	190.27	66.46	256.73
0712	Douple pole structure(cut point) of- 33 kv line using11 m long RSJ pole (152x152)	Nos	15.00	0.89	0.31	13.37	4.67	18.04
0714	Single pole cut point structure for 33kv line on RSJ13 m pole	Nos	86.00	0.84	0.29	72.16	25.20	97.36
New	33kv four pole structure	Nos	9.00	1.96	0.69	17.65	6.17	23.82
New	33KV Isolator / AB Switch on 4Pole OR 2Pole STRUCTURE (cut point) OF- 33 KV LINE	Nos	22.00	0.89	0.31	19.57	6.84	26.41
905	11kv, Pin type with 100 Sq. mm AAAC lines on 100 X 116 mm 11 mtr RSJ poles	KM	1.13	8.7148	3.04	9.8	3.44	13.29
906	11kv, Pin type with 100 Sq. mm AAAC lines on 100 X 116 mm 10 mtr RSJ poles	KM	80.20	6.95	2.43	557.47	194.73	752.20
0912	11kv, Express/Highway Crossing with 100 sqmm AAAC conductor on 152X152MM, 13 mtr RSJ Poles DP with one span of 30 mtr.	Nos	13.00	1.86	0.65	24.16	8.44	32.60
0913	11kv, 3 core X 300 sqmm XLPE Underground Cable	KM	340.80	17.43	6.09	5941.18	2075.25	8016.43
915	11kv 3 core x185sqmm XLPE ug cable	KM	1.00	14.33	5.01	14.33	5.01	19.34
0916	11kv, 3 core X 95 sqmm XLPE Underground Cable	KM	30.00	10.64	3.72	319.30	111.53	430.83
0934R	Upgradation by 11 KV 100 Sq.mm Covered Conductor	KM	60.00	13.19	4.61	791.24	276.38	1067.62

Activity Number	Description of Activity	Unit	NZ Total-Qnty	Tender Cost (Amount in . Lakhs)				
				Part 1: Cost of Material Unit Rate	Part 2: Cost of Services Unit Rate	Total Cost of Material	Total Cost of Services	Total Cost of Services NZ
(1)	(2)	(3)	(4)	(5)	(6)	(8)='(4) x '(5)	(9)='(4) x '(6)	(10) = (9) + (8)
New-HTLS	Upgradation by 33 KV 232 Sq.mm HT LS Conductor	KM	27.00	16.69	5.83	450.66	157.42	608.07
0920	Douple pole structure(cut point) of- 11 kv line using13 m long RSJ pole	KM	20.00	0.97	0.34	19.32	6.75	26.07
0921	Douple pole structure(cut point) of- 11 kv line using11 m long RSJ pole	No.s	106.00	0.88	0.31	92.93	32.46	125.39
0922	Douple pole structure(cut point) of- 11 kv line using11 m long RSJ pole	Nos	47.00	0.69	0.24	32.40	11.32	43.72
0923	Single pole cut point structure for 11kv line on RSJ 9 m pole	No.s	93.00	0.33	0.11	30.24	10.56	40.81
0924	SINGLE POLE CUT POINT STRUCTURE FOR 11KV LINE ON RSJ 11 M POLE	No.s	4.00	0.45	0.16	1.79	0.62	2.41
1203	100 kVA Dist. Transformer centers on 9mtr RSJ poles with MCCB Dist box	No.s	40.00	2.55	0.89	102.14	35.68	137.82
1205	200 kVA Dist. Transformer centers on 11 mtr RSJ poles with MCCB Dist box	No.s	100.00	4.17	1.46	416.70	145.55	562.25
1207	315 kVA Dist. Transformer centers on 11 mtr RSJ poles with MCCB Dist box	No.s	25.00	8.43	2.94	210.64	73.58	284.22
1227	11/0.44 DTC new 315 KVA, Indoor type DTC with RMU	No.s	12.00	12.58	4.39	150.91	52.71	203.62
1406	3.5 X 185 sqmm XLPE UG Cable	KM	8.00	7.31	2.55	58.45	20.42	78.87
1407	3.5 X 300 Sq mm Underground LT XLPE Armoured Cable	KM	14.00	10.45	3.65	146.25	51.09	197.34
1408	3.5 X 240 sqmm XLPE UG Cable	KM	5.00	9.12	3.18	45.58	15.92	61.50
New	Fault passage indicator (FPI)	Nos	200.00	1.00	0.35	200.00	69.86	269.86
New	Replacement of Outdoor switchgear, 33 kV, 1600 Amp	No.s	14.00	2.48	0.60	34.70	8.40	43.10
5740	Replacement of Indoor switchgear, 11 kV, 250 MVA with 2 I/C + 8 OG + 1 BC	Set	4.00	39.43	9.54	157.73	38.17	195.90
5745	33 kV Isolator ,800 A without EB	No.s	6.00	0.64	0.15	3.82	0.92	4.74
6203-R1	Activity for New RMU (SF6 Type) with 2 Isolator + 2 Breaker	Nos	115.00	7.43	2.60	854.41	298.45	1152.86
6203-R2	Activity for New RMU (SF6 Type) with 3 Isolator + 2 Breaker	No.s	42.00	8.87	3.10	372.53	130.12	502.65
2002	Supply erection ,testing & Commissioning of LT 4 way F.P.	No.s	70.00	0.38	0.13	26.49	9.25	35.74
2003	Supply erection ,testing & Commissioning of LT Mini F.P.	No.s	60.00	0.18	0.06	11.02	3.85	14.87
6723	11 kV A.B. Switch 400 Amp with DP structure.	Nos	63.00	0.59	0.21	37.41	13.07	50.48
Total Cost in Lakhs						29858.24	9758.03	39616.27

Note

The type of Insulators considered for the above activities are presumed to be Porcelain type.
In case, the polymers Insulators are to be used, the rates will be as below:

Material Type	Rates (Rs)
11 KV Disc Insulator 70 KN B&S Polymer	142
33 KV Disc Insulator (B&S Polymer)	210
11 kV Pin Insulator Polymer	350
33 KV Pin Insulator Polymer	225

(Results Based Lending India 50193-003)

Appendix 1 to EMP
For Distribution System Strengthening

Government of India National Environmental Standards

Table 1. National Ambient Air Quality Standards and WHO Guidelines

Parameter	Location ^a	National Ambient Air Quality Standards ^b	WHO Air Quality Guidelines ($\mu\text{g}/\text{m}^3$)		Applicable Per ADB SPS ($\mu\text{g}/\text{m}^3$) ^e
			Global Update 2005 ^c	Second Edition 2000 ^d	
Particulate Matter PM_{10} ($\mu\text{g}/\text{m}^3$)	Industrial	60 (Annual)	20 (Annual)	-	20 (Annual)
	Residential, Rural and Other Areas	100 (24-hr)	50 (24-hr)		50 (24-hr)
	Sensitive Area	60 (Annual) 100 (24-hr)	20 (Annual) 50 (24-hr)	-	20 (Annual) 50 (24-hr)
Particulate Matter $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$)	Industrial	40 (Annual)	10 (Annual)	-	10 (Annual)
	Residential, Rural and Other Areas	60 (24-hr)	25 (24-hr)		25 (24-hr)
	Sensitive Area	40 (Annual) 60 (24-hr)	10 (Annual) 25 (24-hr)		10 (Annual) 25 (24-hr)
Sulfur Dioxide SO_2 ($\mu\text{g}/\text{m}^3$)	Industrial	50 (Annual)	20 (24-hr)	-	50 (Annual)
	Residential, Rural and Other Areas	80 (24-hr)	500 (10-min)		20 (24-hr) 500 (10-min)
	Sensitive Area	20 (Annual) 80 (24-hr)	20 (24-hr) 500 (10-min)	-	20 (Annual) 20 (24-hr) 500 (10-min)
Nitrogen Dioxide NO_2 ($\mu\text{g}/\text{m}^3$)	Industrial	40 (Annual)	40 (Annual)	-	40 (Annual)
	Residential, Rural and Other Areas	80 (24-hr)	200 (1-hr)		80 (24-hr) 200 (1-hr)
	Sensitive Area	30 (Annual) 80 (24-hr)	40 (Annual) 200 (1-hr)	-	30 (Annual) 80 (24-hr) 200 (1-hr)
Carbon Monoxide CO ($\mu\text{g}/\text{m}^3$)	Industrial	2,000 (8-hr)	-	10,000 (8-hr)	2,000 (8-hr)
	Residential, Rural and Other Areas	4,000 (1-hr)		100,000 (15-min)	4,000 (1-hr) 100,000 (15-min)
	Sensitive Area	2,000 (8-hr) 4,000 (1-hr)	-	10,000 (8-hr) 100,000 (15-min)	2,000 (8-hr) 4,000 (1-hr) 100,000 (15-min)
Ozone (O_3) ($\mu\text{g}/\text{m}^3$)	Industrial	100 (8-hr)	100 (8-hr)	-	100 (8-hr)
	Residential, Rural and Other Areas	180 (1-hr)			180 (1-hr)
	Sensitive Area	100 (8-hr) 180 (1-hr)	100 (8-hr)	-	100 (8-hr) 180 (1-hr)
Lead (Pb) ($\mu\text{g}/\text{m}^3$)	Industrial,	0.5 (Annual)	-	0.5 (Annual)	0.5 (Annual)
	Residential, Rural and Other Areas	1.0 (24-hr)			1.0 (24-hr)
	Sensitive Area	0.5 (Annual) 1.0 (24-hr)	-	0.5 (Annual)	0.5 (Annual) 1.0 (24-hr)
Ammonia (NH_3) ($\mu\text{g}/\text{m}^3$)	Industrial	100 (Annual)	-		100 (Annual)
	Residential, Rural and Other Areas	400 (24-hr)			400 (24-hr)
	Sensitive Area	100 (Annual) 400 (24-hr)	-	-	100 (Annual) 400 (24-hr)
Benzene (C_6H_6) ($\mu\text{g}/\text{m}^3$)	Industrial	5 (Annual)	-	-	5 (Annual)
	Residential, Rural and Other Areas				

INTERNAL. This information is accessible to ADB Management and Staff. It may be shared outside ADB with appropriate permission.

	Sensitive Area	5 (Annual)	-	-	5 (Annual)
Benzo(o)	Industrial	1 (Annual)	-	-	1 (Annual)
Pyrene (BaP)	Residential, Rural and Other Areas				

Parameter	Location ^a	National Ambient Air Quality Standards ^b	WHO Air Quality Guidelines (µg/m ³)		Applicable Per ADB SPS (µg/m ³) ^e
			Global Update 2005 ^c	Second Edition 2000 ^d	
(ng/m ³)	Sensitive Area	1 (Annual)	-	-	1 (Annual)
Arsenic (As) (ng/m ³)	Industrial	6 (Annual)	-	-	6 (Annual)
	Residential, Rural and Other Areas				
	Sensitive Area	60 (Annual)	-	-	60 (Annual)
Nickel (Ni) (ng/m ³)	Industrial	20 (Annual)	-	-	20 (Annual)
	Residential, Rural and Other Areas				
	Sensitive Area	20 (Annual)	-	-	20 (Annual)

^a Sensitive area refers to Ecologically sensitive areas notified by the India Central Government

^b http://cpcb.nic.in/uploads/National_Ambient_Air_Quality_Standards.pdf

^c WHO Air quality guidelines for particulate matter, ozone, nitrogen dioxide and sulfur dioxide. *Global update 2005*. WHO. 2006.

^d Air Quality Guidelines for Europe Second Edition. WHO 2000.

^e As per ADB SPS, the government shall achieve whichever of the ambient air quality standards is more stringent. If less stringent levels or measures are appropriate in view of specific project circumstances, the executing agency of the government will provide full and detailed justification for any proposed alternatives that are consistent with the requirements presented in ADB SPS.

Table 2. National Noise Standards and WHO Guidelines

Receptor/ Source	Noise Level Standards ^a (dBA)		WHO Guidelines Value For Noise Levels Measured Out of Doors ^b (One Hour LAeq in dBA)		Applicable Per ADB SPS ^c (dBA)	
	Day	Night	07:00 – 22:00	22:00 – 07:00	Day time	Night time
Industrial area	75	70	70	70	70	70
Commercial area	65	55			65	55
Residential Area	55	45	55	45	55	45
Silent Zone	50	40			50	40

^a Noise Pollution (Regulation and Control) Rules, 2002 as amended up to 2010 (<http://cpcb.nic.in/displaypdf.php?id=Tm9pc2UtU3RhbmRhcmRzL25vaXNlX3J1bGVzXzlwMDAucGRm>)

^b Guidelines for Community Noise. WHO. 1999.

^c As per ADB SPS, the project proponent shall achieve whichever of the ambient air quality standards is more stringent. If less stringent levels or measures are appropriate in view of specific project circumstances, the project proponent will provide full and detailed justification for any proposed alternatives that are consistent with the requirements presented in ADB SPS, 2009.

Table 3. National Drinking Water Quality Standards and WHO Guidelines

Group	National Standards for Drinking Water ^{a, b}			WHO Guidelines for Drinking Water Quality, 4th Edition, 2011 ^c	Applicable Per ADB SPS ^{d, e}
	Parameter	Unit	Max. Concentration Limit		
Physical	Turbidity	NTU	1 (5)	-	1 (5)
	pH		6.5 – 8.5	None	6.5 – 8.5
	Color	Hazen Units	5 (15)	None	5 (15)
	Taste and Odor		Agreeable	-	Agreeable
	TDS	mg/l	500 (2,000)	-	500 (2,000)
	Iron	mg/l	0.3	-	0.3
	Manganese	mg/l	0.1 (0.3)	-	0.1 (0.3)
	Arsenic	mg/l	0.01 (0.05)	0.01	0.01
	Cadmium	mg/l	0.003	0.003	0.003
	Chromium	mg/l	0.05	0.05	0.05
	Cyanide	mg/l	0.05	None	0.05

Group	National Standards for Drinking Water ^{a, b}			WHO Guidelines for Drinking Water Quality, 4th Edition, 2011 ^c	Applicable Per ADB SPS ^{d, e}
	Parameter	Unit	Max. Concentration Limit		
Chemical	Fluoride	mg/l	1 (1.5)	1.5	1 (1.5)
	Lead	mg/l	0.01	0.01	0.01
	Ammonia	mg/l	0.5	none established	0.5
	Chloride	mg/l	250 (1,000)	none established	250 (1,000)
	Barium	mg/l	0.7	None	0.7
	Sulphate	mg/l	200 (400)	None	200 (400)
	Nitrate	mg/l	45	50	45
	Copper	mg/l	0.05 (1.5)	2	0.05 (1.5)
	Total Hardness	mg/l	200 (600)	-	200 (600)
	Calcium	mg/l	75 (200)	-	75 (200)
	Zinc	mg/l	5 (15)	none established	5 (15)
	Mercury	mg/l	0.001	0.006	0.001
	Aluminum	mg/l	0.1 (0.3)	none established	0.1 (0.3)
	Anionic Detergents	mg/l	0.2 (1.0)	None	0.2 (1.0)
	Phenolic Compounds	mg/l	0.001(0.002)	None	0.001(0.002)
	Residual Chlorine	mg/l	0.2	5	0.2
Microbial indicator	E-coli	MPN/100ml	Must not be detectable in any 100 ml sample	Must not be detectable in any 100 ml sample	Must not be detectable in any 100 ml Sample
	Total Coliform	MPN/100ml			

^a <http://cgwb.gov.in/Documents/WQ-standards.pdf>.

^b Bureau of India Standard 10500: 2012 (Indian Standard, Drinking Water — Specification (Second Revision)).

^c Health-based guideline values.

^d As per ADB SPS, the government shall achieve whichever of the drinking quality standards is more stringent. If less stringent levels or measures are appropriate in view of specific project circumstances, the executing agency of the government will provide full and detailed justification for any proposed alternatives that are consistent with the requirements presented in ADB SPS, 2009.

^e Figures in parenthesis are maximum limits allowed in the absence of alternate source.

Water quality of surface water bodies is classified by Central Pollution Control Board into five (A, B, C, D and E, see Table 4) quality classes according to their quality suitability for designated best use.

Table 4. Central Pollution Control Board

Designated best use	Quality Class	Primary Water Quality Criteria
Drinking water source without conventional treatment but with chlorination	A	➤ Total coliform organisms (MPN*/100 ml) shall be 50 or less ➤ pH between 6.5 and 8.5 ➤ Dissolved Oxygen 6 mg/l or more, and ➤ Biochemical Oxygen Demand 2 mg/l or less
Outdoor bathing (organized)	B	➤ Total coliform organisms(MPN/100 ml) shall be 500 or less ➤ pH between 6.5 and 8.5 ➤ Dissolved Oxygen 5 mg/l or more, and ➤ Biochemical Oxygen Demand 3 mg/l or less
Drinking water source with conventional treatment	C	➤ Total coliform organisms(MPN/100 ml) shall be 5000 or less ➤ pH between 6 and 9 ➤ Dissolved Oxygen 4 mg/l or more, and ➤ Biochemical Oxygen Demand 3 mg/l or less
Propagation of wildlife and fisheries	D	➤ pH between 6.5 and 8.5 ➤ Dissolved Oxygen 4 mg/l or more, and ➤ Free ammonia (as N) 1.2 mg/l or less
Irrigation, industrial cooling, and controlled disposal	E	➤ pH between 6.0 and 8.5 ➤ Electrical conductivity less than 2250 micro mhos/cm, ➤ Sodium Absorption Ratio less than 26, and Boron less than 2 mg/l.

* MPN: Most Probable Number
(Source: CPCB, 1978)

(Results Based Lending India 50193-003)

**Appendix 2 to EMP
For Distribution System Strengthening**

Daily Monitoring Sheet for Contractors¹

Project Name:
Block Name:
District Name:
MSDCL Division / Sub-Division
Total length of DT line, type (11kV, 33kV) (in km).
Total area of substation (in acres / hectares)
Contract Number:
Date of Monitoring:
Contractor EHS Focal Point Person (C-EHS) Name and Number:

Summary of Findings

Monitoring Item	Status	Remarks
1. Compliance with Local Permit Requirements (Secured / Application Submitted / Not Applicable)		
Location/ Construction permit		
Transport / hauling permits, if any		
List of NOCs obtained or on file		
2. Compliance with EMP Requirements (Approved / Under Preparation / Submitted to PIU for Approval)		
Site EMP (S-EMP)		
Corrective Action Plan, if any		
3. Compliance with S-EMP		
Construction Site (Satisfactory / Needs Improvement / Not Implemented)		
- Conduct of toolbox talk		
- Use of PPE		
Rest areas for male and female workers		
- Toilets for male and female workers		
- Medical kits		
- Drinking water supply		
- Dust control		
- Noise control		
- waste management (all streams)		
- Construction wastewater management		
Temporary Chemicals storage (fuel, oil, etc.)		
- Siltation or erosion control		

¹ This is sample form for use by the EPC Contractors; inspection items will be updated by EPC Contractors in coordination with the PIUs.

Heavy equipment staging / parking
- area

- Barricades around excavation sites

- Access to residential
houses/adjacent properties

- Traffic routing signages

- Lightings at night

Construction Workers' Camp Site	(Available / Needs Improvement / Not Available)	

Quarters for male and female workers

Monitoring Item	Status	Remarks
- Sleeping utilities (e.g. beds, pillows, blankets, mosquito nets, etc.)		
- Power/Electricity supply		
- Drinking water supply		
- Toilets for male and female workers		
General purpose water supply - (cooking, washing, bathing)		
- Cooking facilities and areas		
- Solid waste management		
- Wastewater management		
- Pest control		
4. Implementation of project GRM	(Yes / No or None / Under Resolution)	
Complaints		
Complaints resolution		
5. Environmental Quality Measurement	(Passed / Failed / Not Applicable)	
Ambient air quality sampling		
Noise level measurement		
Surface water quality		

Other Issues: _____

Attachments:

1. Copies of permits secured, if any.
2. Photos taken at worksites, if any.

(Photos attached in previous monitoring sheets should not be used again).

Prepared by: _____

Name, Designation and Signature

Environmental Management Plan

Stage: Draft
September 2024

India: Maharashtra Power Distribution Enhancement
Program for Facilitating Solarization and Expanding
Agricultural Connections

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Appendices

Appendix 1. Government of India National Environmental Standards

Appendix 2. Template for Daily Monitoring Sheet for Contractors

A. INTRODUCTION

1. This Environmental Management Plan (EMP) has been developed for bid, tender and contract purposes by Maharashtra State Electricity Distribution Co. Limited (MSEDCL) for Sub-component 2 under Output 1 as shown:

Output 1: Low carbon development of the agriculture and power sectors through Private Sector investments facilitated.	Sub-component 2 under Output 1: Distribution system strengthening to facilitate agriculture feeder solarization to supply daytime reliable power to agriculture consumers and connect the household rooftop solar system to the electricity grid.
--	--

B. ENVIRONMENTAL MITIGATION AND MONITORING PLAN (EMP)

2. This EMP identifies the potential environmental impacts and defines mitigation measures and monitoring requirements for the design, pre- construction, construction stages related to new construction of 33/11kV substations, 33kV and 11kV distribution lines, system strengthening and augmentation works of existing assets i.e., additional transformer, augmentation at existing substations and reconductoring of existing distribution lines (henceforth, all are referred to as “projects”). These mitigation and monitoring requirements shall be adopted by the Engineering, Procurement, Construction (EPC) Contractors. The EMP will ensure safeguards protection activities during project implementation to prevent, reduce or mitigate potential adverse environmental and social impacts and risks, including Health & Safety risks (Occupational Health & Safety, Community Health & Safety).

3. Preliminary designs and detailed engineering designs will be prepared by the EPC Contractors and approved by the MSEDCL Project Management Unit (PMU); this may result in subsequent revisions to this EMP. The MSEDCL will provide the submitted detailed designs to the Consultants supported by the Asian Development Bank (ADB) for review to determine if the EMP requires revision and update. The draft and final EMP will be disclosed on the public MSEDCL (<https://www.mahadiscom.in/en/home/>) and ADB public website (<https://www.adb.org/>).

4. The project works will be executed by the EPC Contractors on turn-key basis, or package contract to design and build. The contractors¹ will be informed of their obligations to implement the EMP and to include EMP implementation costs in their bids for project works.

C. EMP IMPLEMENTATION RESPONSIBILITIES

5. The EPC Contractors will be responsible for implementing the mitigation and monitoring requirements during design, pre-construction and construction stages; the EPC Contractors will work

¹ Contractors imply = Engineering Procurement, Construction (EPC) contractors, any key sub-contractors. (Note: EMP implementation during operation & maintenance (O&M) stage will continue through the Project Implementation Units (PIUs) set up at MSEDCL Division level (and as per the operational set up within MSEDCL – described in the Program Safeguards System Assessment); and for any strengthening and augmentation works at existing transmission substations through the Maharashtra State Electricity Transmission Company Limited (MSETCL) that is the facility operator for transmission substations across the state of Maharashtra).

in coordination with and be supervised and monitored by MSEDCL Project Implementation Unit (PIU) engineers set up at Division Level. The EPC Contractor will appoint their own Contractor Environment, Health and Safety (C-EHS) focal point person(s) per site who will be responsible for safeguards implementation as per project EMP, Grievance Redressal as per GRM as well as Health and Safety (occupational health and safety, community health and safety). The EPC Contractors will be required to prepare a site-specific EMP (S-EMP); refer to Table 2 of EMP for list of plans. The EPC Contractors will bear the costs of preparing the S-EMP; these will help to avoid unplanned activities of the EPC contractors and will guide the smooth implementation of all project activities.

6. The contractors will not commence any civil works until all No Objection Certificates (NOCs), permits and/or clearances have been granted to the PIUs and PMU / ADB has approved the final EMP based on detailed engineering design. Contractors shall:

- Work with the PIUs to finalize site specific S-EMPs during construction²;
- Appoint focal point person C-EHS per site within its staff on full time basis;
- Attend training and capacity building sessions;
- Conduct orientation and daily briefing sessions to workers on Health and Safety, project GRM;
- Submit site specific EMP (S-EMP) for construction to the PIU for review and approval;
- Submit monthly progress reports to the PIU on the implementation of the S-EMP / EMP;
- Engage in project GRM in co-ordination with the PIU and area Gram Panchayat³ to provide up to date information to relevant stakeholders on project activities and address any grievances, issues that arise during project implementation effectively and in a timely manner;
- Ensure that appropriate worker facilities (workers accommodation / camps) are provided at work sites in line with the project EMP;
- Register and maintain records of all grievances and corresponding resolution processes, work-related accidents, and undertake remedial actions to mitigate/minimize recurrence;
- Implement EMP measures and report to the PIU if any new impacts surface; seek guidance from the PIU as required for S-EMP / EMP implementation;
- Conduct quantitative environmental monitoring (air, noise, etc.) as per requirements in the EMP (Government of India national environmental standards are enclosed as Appendix 1 to EMP
- Conduct site audit one a month during peak construction stage to ensure checks are in place for regulatory compliance (national / state / local), documentation review (such as all permissions / NOCs are on file), Consultations and GRM log-book is maintained, S-EMP/ EMP is implemented; Incident, spills, leaks and rectification log-book is maintained, etc. Site audit including recommendations and action plan for any deviances shall be submitted to the PIU at the end of the month

7. The PIU will ensure that the EPC Contractors are aware of their obligations including specific provisions requiring EPC Contractors to comply with: (i) all applicable labor laws and core labor standards on (a) prohibition of child labor as defined in national legislation for construction activities; construction site should not hire any child below 18 years of age; (b) equal pay for equal work of equal value regardless of gender, ethnicity, or caste including no discrimination against pregnant women; and (c) prohibition of forced labor and with (ii) the requirement to disseminate information

² The S-EMP will need to be fully compliant with the EMP and will need to be prepared within 30 days of contract award and approved 10 days prior to access to the site.

³ Gram Panchayat is a local governing body in Indian villages that is responsible for the administration of a village or group of villages. The term "Gram Panchayat" translates to "village council"

on health & safety risks due to transmittable diseases, including HIV/AIDS and coronavirus disease (COVID-19) to employees.

D. MONITORING AND REPORTING PROCEDURES FOR EMP

8. During construction, the EPC Contractor will prepare the monthly progress reports on S-EMP/EMP implementation and submit to the PIU. The EPC Contractor monthly progress reports will include compilation of daily monitoring sheets that is duly signed by the PIU engineers; a sample monitoring template is enclosed as Appendix 2 to EMP.

E. INFORMATION DISCLOSURE, CONSULTATION AND PARTICIPATION

9. The EPC Contractor in coordination with the PIU and area Gram Panchayat will undertake consultations with any affected persons, households, settlements and other relevant stakeholders residing in the project area. The EPC Contractor in coordination with the PIU will prepare a consultation and participation plan with schedule, location, invited participants, information to be disseminated and methods of consultation. An overview activity outline for a Consultation and Participation is shown below.

10. Consultations with any affected persons, households, settlements, Gram Panchayat members and relevant stakeholders will include dissemination of project information leaflet in the local language (Marathi). The leaflet will include project information such as on project environmental impacts (positive and negative), safeguards measures including health and safety (CHS), risks due to transmittable diseases (HIV/AIDs, COVID-19), construction implementation schedule, stakeholders right to complain and the project GRM. The leaflet will also include the contact information including EPC Contractor address and telephone number for the contractor focal point persons (C-EHS) and PIU engineers.

11. Such consultations will provide a two-way information-sharing channel, ensuring that the concerns, questions and ideas of the local stakeholders will be discussed and responded to in an appropriate and gender inclusive way. The contractors C-EHS, will record all information dissemination and consultation activities and the results from consultations with the local stakeholders. These records will demonstrate how concerns raised, if any, and recommendations made are addressed in the updated EMP. Consultations will be supported and supervised by the PIU engineers and submitted to the PMU. Consultations with affected persons, households, settlements, area Gram Panchayat and other relevant stakeholders will continue throughout project implementation and will be open and gender inclusive.

12. The updated EMP will be disclosed on MSEDCL and ADB website as required by the ADB SPS 2009 and Access to Information Policy (September, 2018)⁴

ACTIVITY OUTLINE FOR CONSULTATION AND PARTICIPATION AND INFORMATION DISCLOSURE

Project Implementation Schedule	Activity	Stakeholders
Detail design phase: Detailed Walkover	Public information meetings Informal meetings on project design for aligning people's responses in detailed design	Affected persons Households

⁴ <https://www.adb.org/documents/access-information-policy>⁵ <https://www.adb.org/site/accountability-mechanism/main>

Project Implementation Schedule	Activity	Stakeholders
Survey and once prior to construction	Dissemination of project information leaflet Community health and safety, and/or awareness program on environment and social safeguards one (1) month prior to civil works, including informal meetings for information updates on project schedule and activities through PIUs MSEDCL GRM	Settlements Gram Panchayat
Civil works construction	Informal meetings for information updates on project schedule and activities through PIUs Public information meetings, as needed Community awareness program on environment and social safeguards one (1) time during civil works Information leaflet made available at public information meetings, PIUs and Section offices MSEDCL GRM	
O&M	Informal meetings for information updates and concerns, as needed	

F. PROJECT GRM

13. MSEDCL does not have a specific project level grievance redress mechanism (GRM) for safeguards although they have established a GRM for the previous RBL loan. MSEDCL as the State Utility, however, has a grievance redress system that is effective, transparent and accessible to the public all the way from HQ to Villages across the State of Maharashtra i.e., at Section level. Currently, all consumers and beneficiaries are listed in the MSEDCL's database and are made aware of their concerned nodal officer(s). MSEDCL has both electronic and manual mode of accepting grievances. The existing utility level GRM is mostly for the operation, maintenance and for the customer complaints and not particularly for any project or program. Efforts will be undertaken to make all beneficiaries and relevant stakeholders aware of the current MSEDCL GRM. For the proposed RBL program (Project), following project level GRM is proposed for project implementation and monitoring.

- i. First tier (Contractor). At field level, any construction related grievances will be registered formally by contacting the Contractor (C-EHS). If the grievance cannot be resolved at first tier, then the contractor can assist affected persons / aggrieved party to take forward the registered complaint to second tier. The contractor must keep records of all grievances received. The contractor will be responsible to resolve the complaint within 7 days and communicate the decision to the affected persons / aggrieved party formally in writing. Note: The affected person / aggrieved party can also directly approach the MSEDCL Internal Grievances Redressal Cell (IGRC) at any time.
- ii. Second Tier (area Grievance Redress Committee) at subdivision level. If or when the affected person / aggrieved party is not satisfied with the action or decision of the contractor, the affected person / aggrieved party will take the grievance to the area Grievance Redress Committee comprising of the Gram

Panchayat (Sarpanch or area representatives) and PIU E&S focal point persons. The Committee will screen the received grievance to determine whether the concerns raised in the grievance are within the scope of the project. The GRC will determine solutions to the issues either by (i) discussing internally, or (ii) joint problem solving with affected person / aggrieved party, or (iii) a combination of both options. If the complaint is resolved within 15 days, the GRC must communicate the decision to the affected person /aggrieved party formally in writing. Should matter be unresolved and/or the affected persons / aggrieved party are not satisfied with the resolution, the complaint will be referred by the GRC to the third tier.

- iii. Third Tier (PIU Level/Division Level). If the issue/complaint cannot be resolved in a way that is satisfactory to all parties, the GRC will bring the grievance to the PIU at the Division Office. The PIU Division engineer will meet the affected person / aggrieved party and GRC and try to resolve the issue/ grievance. Within 15 days of the submission of the complaint to the PIU level, it has to take a decision and inform in writing to the affected person / aggrieved party and GRC of the decision.
- iv. Fourth Tier (PMU Level/HQ level). If the issue / complaint cannot be resolved in a way that is satisfactory to all parties, the PIU (Division engineer) and PMU (at HQ) will hold discussions on the matter internally and if necessary, arrange visit to the project site and hold on-site discussions with the affected person / aggrieved party, GRC and contractor C-EHS. Within seven days of the submission of the complaint to the PMU, it has to take a decision and inform in writing both the affected person / aggrieved party and GRC of the decision.

14. Access to ADB. If efforts to resolve disputes using the GRM remain unresolved or unsatisfactory, the affected person / aggrieved party have the right to directly discuss their concerns or problems with the ADB project officer at ADB HQ. The complaint may be submitted in any of the official languages of ADB DMC. This may be done at any time sending the written complaint to the following address. If affected person / aggrieved party is still not satisfied with the responses of the ADB project officer, they can directly contact the ADB Office of the Special Project Facilitator. The Office of the Special Project Facilitator procedure can proceed based on the accountability mechanism⁵ in parallel with the project implementation.

Project officer: Maharashtra Power Distribution Enhancement Program for Facilitating Solarization and Expanding Agricultural Connections

Energy Sector Group
Asian Development Bank
6 ADB Avenue, Mandaluyong City
1550 Metro Manila, Philippines

15. Country Legal System. The GRM notwithstanding, an affected person / aggrieved party shall have access to the country's legal system at any stage through Government of India's judicial or appropriate administrative system. This can run parallel to accessing the project GRM and is not dependent on the negative outcome of the project GRM.

⁵ <https://www.adb.org/site/accountability-mechanism/main>

16. **GRC Recordkeeping and Reporting.** The EPC Contractor and the PIU will keep a record of all the grievances received including contact details of complainant, date the complaint was received, nature of grievance, agreed corrective actions, the date these were affected and the final outcome. Documentation of the grievances filed and resolved will be summarized, submitted to the PMU (and reported in the semi-annual progress reports, safeguards section, during project implementation).

17. **Disclosure of Information.** Under the direction of the PIU, the EPC Contractors will disseminate project information leaflet and also inform affected persons / aggrieved party / relevant stakeholders on grievance redress procedure, who to contact and when, where and how to file a grievance, time likely to be taken for redressal of minor and major grievances, etc. It will be made known that affected persons / aggrieved party are not charged any fee under project GRM. It is a good practice to provide the affected person /aggrieved party with transport facilities to the GRC hearings, if required. Grievances received and responses provided will be documented and provided to affected persons / aggrieved party during the process. The number of grievances recorded and resolved and the outcomes will be displayed/disclosed in the project construction field offices.

18. **Review of the Process.** The PIU will periodically review the implementation of the project GRM and record information on the effectiveness of the mechanism, especially on the project ability to prevent and address grievances.

G. MECHANISMS FOR FEEDBACK AND ADJUSTMENT

19. The EMP is a living document. The need to update and adjust the EMP will be reviewed during final project detailed engineering design or when there are design changes, unfavorable environmental monitoring results or inappropriate monitoring locations and ineffective or inadequate mitigation measures. Based on environmental monitoring and reporting systems in place, the PIU shall assess whether further mitigation measures are required as corrective action, or improvement in environmental management practices are required. The PIU will inform PMU promptly on any changes to the project and needed adjustments to the EMP. The updated EMP will be submitted to PMU for review and approval and will be disclosed on the MSEDCL website.

H. COST ESTIMATES

20. There are several types of mitigation measures and corresponding costs:

- i. Measures that will permanently become part of the infrastructure such as landscaping, re-seeding of project site if applicable, hedge planting, maintenance of perimeter margins, fencing / wall, buildings / road signage, detailed engineering measures for preventing soil erosion, defining and strengthening natural drainage pattern, climate resilience measures will be included within the main civil works contract cost and not double-counted as part of the EMP costs.
- ii. Orientation of contracted workers per site can be counted as part of the contractor costs.
- iii. Measures during pre-construction (such as conducting quantitative environmental monitoring to establish baseline) will be included in the tender documents to ensure that all contractors budget these as a specific line item in their bids, including procurement of monitoring equipment such as hand-held portable air and noise monitoring devices, that will be the property of the contractor.

- iv. Measures during the construction stage (e.g., periodic watering for dust suppression, use of hand-held portable air and noise monitoring devices, low noise machinery and equipment, PPE, etc.), as well as measures to mitigate unforeseen impacts due to construction activities will need to be included in the tender documents to ensure that all contractors budget these as a specific line item in their bids.
- v. Information disclosure, developing project information leaflet, GRM related costs involved in resolving the complaints (meetings, consultations, communication and reporting/information dissemination) will be borne by the PMU / MSEDCL.

I. EMP TABLES

See next page

(Results Based Lending India 50193-003)

ENVIRONMENTAL MANAGEMENT AND MONITORING PLAN FOR EPC CONTRACTORS

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)						
Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
		Site Selection⁸				
Impact on Valued Environmental Receptors (VERs)	A.1	Site selected for new S/S avoids Valued Environmental Receptors (VERs) such as households / settlements / structures, Physical Cultural Resources (PCRs ⁹), surface water sources / bodies, groundwater wells, bunds, drainage, canals, dense tree clusters ¹⁰ , etc	New S/S and wider area of influence	Walk-over surveys completed, maps used, GPS coordinates of VERs marked, and site boundaries adjusted	Survey results and adjusted boundaries; reviewed by PIU, approved from PMU	Once
	A.2	Site selected for new S/S shall be at least 50 m away from the nearest household / settlement	new S/S	Surveys completed and site boundaries adjusted	Survey results and adjusted boundaries; reviewed by PIU, approved from PMU	Once
	A.3	Detailed walkover / drone survey for new DT line avoids VERs; pole placement in or in close proximity to water bodies avoided; adjusting ROW along road easement / reserves as much as	DT lines, ROW and wider area of influence	Transect surveys completed, maps used, GPS coordinates of VERs marked and ROW adjusted	Survey results and adjusted transect / ROW; reviewed by PIU, approved from PMU	Once

⁶ S/S = Substation; DT = Distribution Line; ROW = Right of Way; Project defined area of influence = new substation or existing substation and buffer of 50 m around the substation, new distribution line or existing distribution line and a buffer of 25 m on each side of the Right of Way center-line

⁷ PIU = Project Implementation Unit of MSEDCL at Division Level (Asst Engineer, Infrastructure + Asst Engineer, Civil for all Safeguards and Dy Executive Engineer, specifically for Occupational Health and Safety); PMU = Project Management Unit at Headquarters, Mumbai

⁸ The PMU/PIUs will conduct the site selection exercise based on the Key Exclusion Criteria for project selection under RBL prior to hand over of selected site to EPC Contractors; the Key Exclusion Criteria is enclosed as Annex 1 of the Program Safeguards System Assessment

⁹ PCRs = local heritage sites / archaeological sites, places of worship

¹⁰ Placement of new S/S or DT ROW through dense tree clusters will be avoided; trees below 3 m will not be cut e.g., along DT ROW; If tree cutting is required, information will be maintained for tree type + tree height + tree count / compensation ratio and offset options will be considered to avoid loss of indigenous, fruit bearing trees. Prior permission for tree cutting will be obtained by the PIU from the competent authority; a copy of granted permission will be maintained at EPC contractor field office. Compensation will be paid in a timely manner to affected persons.

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)						
Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
		possible ¹¹				
		Design for Effective Environmental Management				
Lack of sufficient design and planning to ensure long term sustainability of project and protection of assets created	B.1	Design to consider site elevation and/or new S/S building elevation and pole placement in response to anticipated climate change risks such as flooding	new S/S + DT lines, ROW	Detailed design includes measures B.1, Table 1	Designs approved from PMU; work plans prepared and approved by PIU	Once
	B.2	Design to integrate seismicity considerations for the given geography in Maharashtra (MH) as per National guidance on seismic design that calls for identification of a maximum credible earthquake scenario and associated ground acceleration parameters	new S/S + DT lines, ROW	Detailed design includes measures B.2, Table 1	same as above	Once
	B.3	Design to integrate recommendations from relevant detailed site investigations (geotechnical, topographical, hydrological) and include provisions for ensuring effective Operation & Maintenance (O&M)	new S/S + DT lines, ROW	Detailed design includes measures B.3, Table 1	same as above	Once
	B.4	Design of new S/S will enable efficient drainage of site and maintain natural drainage pattern; or planned drainage works (if any) designed to utilize and maintain natural drainage patterns and to blend in the environment	new S/S	Detailed design includes measures B.4, Table 1	same as above	Once
	B.5	Design to include permanent provision for adequate fencing/barricade /wall and lights around new S/S	new S/S	Detailed design includes measures B.5, Table 1	same as above	Once

¹¹ ROW will be planned along road easements/ reserves as much as possible. In case of unavoidable impacts on crops, trees and other assets, the same will be compensated in a timely manner to affected persons.

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)

Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
	B.6	Design to include septic tank of adequate capacity for permanent sanitation facilities at new S/S building	new S/S	Detailed design includes measures B.6, Table 1	same as above	Once
	B.7	Design a permanent (bund) impermeable surface and/or dykes capable of carrying 110% volume of materials for accidental spills or leakage e.g., for transformer oil drums (sited away from direct sunlight / rain and at least 10 m from water sources, bodies, drainage, bunds)	new S/S + existing S/S	Detailed design includes measures B.7, Table 1	same as above	Once
	B.8	Design to integrate an oil absorbent / containment system e.g., power transformer pad	new S/S + existing S/S	Detailed design includes measures B.8, Table 1	same as above	Once
	B.9	DT ROW design to consider suitable slope and soil type; conduct soil quality testing as part of the engineering scope and check where pole footings require extra reinforcement	DT lines, ROW	Detailed design includes measures B.9, Table 1	same as above	Once
		Planning, Procurement, Logistics				
Selection /List of materials and/or equipment	C.1	Selection / List of materials and/or equipment to conform to MSEDCL Material Technical Specifications ¹² and provisions listed in this EMP ¹³	new S/S + existing S/S, DT lines, ROW	Selection / List of equipment is part of procurement package	Selected equipment approved by PIU	Once
Selection of transformers, electrical equipment (potential release of toxic chemicals)	C.2	Polychlorinated Bi-phenyls (PCBs) shall not be used in new transformers and other electrical equipment refer to Regulation of Polychlorinated Biphenyls	new S/S + existing S/S, DT lines, ROW	Part of tender specifications	Measures integrated into contract document	Once

¹² <https://www.mahadiscom.in/supplier/en/material-technical-specifications/>

¹³ MSEDCL has approved list of vendors (for all schemes), weblink: <https://www.mahadiscom.in/supplier/en/approved-vendor-list-all-schemes/>

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)

Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
		Order, 2016 ¹⁴ .				
	C.3	Equipment, systems shall not use CFCs, including halon; refer to GOI requirements ¹⁵	new S/S + existing S/S, DT lines, ROW	Part of tender specifications	Measures integrated into contract document	Once
Selection of S/S type (potential release of fugitive GHG emissions)	C.4	Design Air Insulated S/S to (AIS) avoid fugitive emissions of sulfur hexafluoride (SF6), a potent Greenhouse Gas (GHG)	new S/S	Part of tender specifications	Measures integrated into EPC contract document	Once
Sources of construction material, re-use of earth cut materials	C.5	Identify sources, vendors of construction materials; specify materials that are recycled, have recycled content or are from sustainable sources; materials to be obtained from government approved vendors	new S/S + Existing S/S, DT lines, ROW	List of sources, vendors	Contracts signed with vendors; records maintained	Once
Sources of water for construction / O&M	C.6	Groundwater abstraction for construction to be avoided; bulk water to be drawn from existing water supply source with due permission from relevant authority obtained by PIU prior to any project activity. If abstraction is unavoidable, then due permission from relevant authority to be obtained by PIU prior to any project activity	new S/S	Part of contract clause	Copy of contract	Once

¹⁴ [Government of India Regulation of PCBs Order, 2016](#)

¹⁵ [Government of India, Ozone cell](#)

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)

Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
Identification of borrow areas, spoils area	C.7	Identify and use borrow areas / spoils areas licensed by the relevant local authority; obtain prior permission from the relevant authority for use ¹⁶	area of influence	List of borrow / spoils areas	Borrow area / spoils site identified; and due approval for use obtained from relevant authorities	Once
Identification of waste / hazardous waste vendors	C.8	Identify and contract Licensed waste / hazardous waste vendors for collection, transfer, treatment and/or final disposal of waste / hazardous waste and/or materials ¹⁷	area of influence	List licensed waste / hazardous waste vendors	Contracts signed with vendors; memorandum of understanding (MOUs) with licensed depots, disposal sites, scrap yards	Once
		Quantitative Environmental Surveys				
Surface water pollution due to run-off from general construction activities, spills etc	D.1	Surface water quality monitoring once before the start of civil works to establish baseline	any natural receiving surface water source/ body that is immediately adjacent to work sites and/or downstream of work sites, if any	Sampling and chemical analysis	Surface water quality monitoring conducted	Once prior to start of civil works

¹⁶ Measures in C.7, Table 1 to be undertaken in coordination with the PIU¹⁷ Measures in C.8, Table 1 to be undertaken in coordination with the PIU

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)						
Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
Air quality and increase in fugitive dust due to civil works, movement of vehicles, heavy machinery	D.2	Air quality monitoring once before the start of civil works to establish baseline	within work site, and sensitive human receptors (households /settlements / schools, etc) downwind of work sites, if any	PM10, PM2.5, SOx, NOx, CO	Monitoring conducted using hand held / portable air monitoring devices	Once prior to start of civil works
Noise due to civil works	D.3	Noise level measurements once before the start of civil works to establish baseline	within work site, and sensitive human receptors (households /settlements / schools, etc) downwind of work sites, if any	Noise (dB) level measurements	Monitoring conducted using hand held / portable devices	Once prior to start of civil works
		Information Disclosure, Stakeholder Consultations and Participation¹⁸				

¹⁸ Measures listed under Information Disclosure, Stakeholder Consultation and Participation E.1 through E.3, Table 1 to be undertaken in coordination with the PIU

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)						
Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
Limited outreach and consultation	E.1	Schedule activities throughout project implementation to engage with stakeholders on topics related to safeguards such as (but not limited to): project design, implementation schedule, key construction activities (in particular that results in disturbance or nuisance), community health and safety (CHS) issues and safety training for communities residing close to new S/S, DT ROW, project Grievance Redress Mechanism (GRM), compensation procedures	area of influence	Announcement to the communities of civil works schedule	Consultation log-book; Leaflets distributed	Once
	E.2	Consult with relevant stakeholders for sensitive periods for religious / spiritual activities	area of influence	Consultation conducted	Consultation log-book; Sensitive periods identified	Once
	E.3	A project information leaflet in Marathi to be prepared and made publicly available in area Gram Panchayat office, MSEDCL Section office (village level) and PIU	area of influence	Preparation of project information leaflet in Marathi	Leaflets publicly available at offices	Once prior to start of civil works
		Project Grievance Redress Mechanism (GRM) / Grievance Redress Committee (GRC)¹⁹				
Smooth project implementation affected due to lack of stakeholder engagement	F.1	Establish and operate an efficient and functional GRM / GRC	area of influence	project GRM / GRC	GRM / GRC set up	Once

¹⁹ Measures listed under Project GRM / GRC F.1 through F.3, Table 1 to be undertaken in coordination with the PIU

Table 1. Design and Pre-Construction Stage: All Infrastructure under Distribution System Strengthening a to facilitate Agriculture Feeder Solarization (sub-component under Output 1)						
Key Activities / Potential Impacts & Issues during design & pre-construction	No	Mitigation Measures and Safeguards	Monitoring Location ⁶	Monitoring Parameter	Method of Measurement ⁷	Frequency
Potential interference to existing utilities such as existing irrigation canals, drainage, bunds, water pumps, groundwater wells, damaged roads, electricity outages, grazing sites, etc	F.2	Resolve queries, conflicts and complaints, if any, in a timely manner.	area of influence	project GRM / GRC	GRM log-book	Ongoing basis
	F.3	Issue public notices via project information leaflet to inform the public of the nearest village, Gram Panchayat with respect to the project defined area of influence of the GRM procedures and contact information of responsible persons	area of influence	distribution of leaflets	Public notices issued; leaflets distributed	One month prior to start of civil works
		Environmental Management and Monitoring Plan (EMP) Updating				
Non-compliance with Asian Development Bank Safeguards Policy Statement (ADB SPS) 2009 and/or the Government of India National environmental and social regulatory framework	G.1	Update EMP based on final project detailed engineering design	area of influence	Updating EMP	Final detailed engineering design incorporated; EMP updated	Once
	G.2	Submit revised documents to PIU / PMU for approval and disclosure on MSEDCL website when updated	area of influence	Updating EMP	Obtain approval from PIU/ PMU and disclosure on MSEDCL website	Once

Table 2. Key Activities by EPC Contractors Prior to Mobilization on Work Site						
Key Activities	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁰	Monitoring Parameter	Method of Measurement	Frequency
		Orientation, Hiring and Presence of Construction (Workers) Camps				
Orientation / awareness of contractors and workers on the safeguard requirements and their responsibility during implementation	A.1	Contractor to conduct briefing and orientation of workers on site specific EMP (S-EMP) implementation, use of Personal Protective Equipment (PPE), Occupational Health and Safety (OHS) risks, hygiene at work sites, socially transmittable diseases to prevent potential incidences	area of influence	Worker's orientation	Record and maintain orientation log-book with list of participants, duration and orientation exercise	Once prior to mobilisation on site
Hiring of workers: Avoid conflict due to workers migration to project area; lack of local support to the project; dispute over transparency of hiring	A.2	Contractors required to use local labor for manual work to avoid conflict due to workers migration; maintain transparency in hiring	area of influence	Local recruitment drive	Maintain a record of number of local workers recruited	Once prior to mobilisation on site
Presence of Construction (workers) camps and workers at work sites	A.3	Clear demarcation of construction (workers) camps in a pre-designated area (within work sites), with vegetation clearance limited to its footprint and at least 10 m from water sources / bodies, drainage; accessible through an existing approach road	new S/S + DT lines, ROW	Construction (workers) camp site demarcation	Camps clearly demarcated; 10 m distance maintained	Once prior to mobilisation on site
	A.4	If located outside of work site, consult with area Gram Panchayat when selecting and placement of camp; at least 10 m from water sources / bodies, drainage	new S/S + DT lines, ROW	Construction (workers) camp site demarcation	Camps clearly demarcated; 10 m distance maintained	Once prior to mobilisation on site

²⁰ S/S = substation; DT = distribution line; ROW = Right of Way; Project defined area of influence = new substation or existing substation and buffer of 50 m around the substation, new distribution line or existing distribution line and a buffer of 25 m on each side of the right of way center-line

Table 2. Key Activities by EPC Contractors Prior to Mobilization on Work Site						
Key Activities	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁰	Monitoring Parameter	Method of Measurement	Frequency
	A.5	Construction (workers) camps to have adequate facilities for temporary stay such as electricity, portable water, portable toilets, waste collection area, first aid kit, etc.; camp to be in line with IFC (WBG) ²¹ Guidance Note / Workers Accommodation ²²	new S/S	Provisions for workers stay in compliance with IFC Guidance Note	Ocular inspection	Once prior to mobilisation on site
		Preparation of Construction Implementation Schedule and Site specific EMP				
Prepare Construction Implementation Schedule	B.1	Contractor to prepare and submit construction implementation schedule; schedule project activities in dry season, avoid sensitive periods for aves (breeding), cropping season, bird/bat roosting sites, days of religious / spiritual activity	area of influence	Part of contract clause	Measures B.1, Table 2 included in Construction Implementation Schedule; Schedule submitted to PIU/PMU is approved	Once prior to mobilisation on site
;	B.2	Contractors to include in the Construction Implementation Schedule estimates of anticipated annual water usage for civil works / construction and sources of water supply	new S/S + DT lines (pole)	Part of contract clause	Measures B.2, Table 2 included in Construction Implementation Schedule; Schedule submitted to PIU/PMU is approved Memorandum of Understanding (MOU) between contractor and service provider obtained and kept on file	Once prior to mobilisation on site

²¹ IFC (WBG) = International Finance Corporation (World Bank Group)

²² <https://www.ifc.org/content/dam/ifc/doc/mgrt/workers-accommodation.pdf>

Table 2. Key Activities by EPC Contractors Prior to Mobilization on Work Site						
Key Activities	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁰	Monitoring Parameter	Method of Measurement	Frequency
Prepare S-EMP to avoid effects of unplanned activities & smooth safeguards implementation	B.3	Contractor to develop site specific EMP (S-EMP) with maps with clearly identified areas; Map will show each work site location with GPS coordinates including at minimum: Access routes, storage areas for waste, materials, equipment, chemicals (such as oils or fuels), re-fuelling locations for vehicles, if any, stockpile areas (on & off site), first aid kit and equipment used in emergency response, location of construction (worker) camps, location of UG utilities (on & off site), if any in the vicinity, location of temporary adequate receiving body for construction run-off List of plans:	new S/S + existing S/S, DT lines, ROW	Part of contract clause	S-EMP prepared and submitted to PIU/PMU is approved	Once prior to mobilisation on site
land and vegetation	B.3.1	Spoils Disposal				
	B.3.2	Site Rehabilitation and Clean-up				
Air	B.3.3	Dust Control				
Noise	B.3.4	Construction Noise				
Water	B.3.5	Construction Wastewater				
material resources	B.3.6	Materials Management Plan (including warehouses / storage)				
Waste	B.3.7	Hazardous Materials Control (including spills response and clean-up)				
Waste	B.3.8	Waste management (for all waste streams) - for handling, storage, transport and disposal of wastes and hazardous materials by licensed vendors approved by MSEDCL				
workers / communities	B.3.9	Community Health and Safety (CHS)				
workers / communities	B.3.10	Occupational Health and Safety (OHS)				
workers / communities	B.3.11	Emergency preparedness and response				

Table 2. Key Activities by EPC Contractors Prior to Mobilization on Work Site						
Key Activities	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁰	Monitoring Parameter	Method of Measurement	Frequency
PCRs	B.3.12	Chance Find Procedures for Physical Cultural Resources (PCRs)				
		Monitoring Checklists				
Site specific checklists to use in quantitative environmental monitoring	C.1	Checklists to show monitored environmental data with data collection points (GPS coordinates) marked on site maps	new S/S + DT lines, ROW	Checklists prepared and submitted to PIU for review and approval	Approved checklists by PIU	Once prior to mobilisation on site
		Information Disclosure, Stakeholder Consultations and Participation²³				
Information Disclosure, Consultation and Participation	D.1	Inform and consult with local residents and area Gram Panchayat; distribute project information leaflet	area of influence	Consultations conducted	Consultation log-book; leaflets distributed	Once prior to mobilisation on site

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
		Local Ecology				
Impact on habitat, species due to site preparation + civil works	A.1	Bird / bast roosting sites, if any found, will be marked and not disturbed at work sites	new S/S + existing S/S	Roosting sites marked	Ocular inspection	Once before start of site preparation + construction

²³ Measures under Information Disclosure, Stakeholder Consultations and Participation D.1, Table 2 to be undertaken in coordination with the PIU

²⁴ S/S = substation; DT = distribution line; ROW = Right of Way; Project defined area of influence = new substation or existing substation and buffer of 50 m around the substation, new distribution line or existing distribution line and a buffer of 25 m on each side of the right of way center-line

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	A.2	Schedule activities in dry season and avoid sensitive periods (breeding periods for aves)	area of influence	Breeding period of aves ascertained	Project work scheduled for dry season and/or sensitive periods of breeding for aves avoided	Once before start of site preparation + construction
		Land and Vegetation				
Impact on land and vegetation due to site preparation + civil works	B.1	Clear demarcation of new S/S boundaries with temporary barricades or fencing; no encroachment outside demarcated area /work sites	new S/S	Clear demarcation of new S/S site; barricades / fencing set up	Ocular inspection / spot checks	Once before start of site preparation + construction
	B.2	Land / vegetation clearance in demarcated work sites; hedges and field margins will be retained as appropriate	new S/S	Land / vegetation clearance	Ocular inspection / spot checks	Once before start of site preparation + construction
Impact on Valued Environmental Receptors (VERs)	B.3	Confirm and verify VERs are avoided; tree cutting avoided and/or due permission from competent authority obtained and on file	new S/S	VERs; tree type + tree height + tree count / compensation ratio and offset options	VERs avoided; Tree cutting permission on file; records maintained	Once before start of site preparation + construction
	B.4	Additional transformer work, augmentation works - to take place within the existing S/S fenced boundary	existing S/S	Clear demarcation within fenced existing S/S boundary	Ocular inspection	During site preparation + construction
Use of pesticides and herbicides	B.5	Use of pesticides or herbicides for land / vegetation clearance is prohibited; employ use of manual or machine clearance	new S/S + existing S/S	Vegetation clearance method by manual or machine clearing	Ocular inspection / spot checks	During site preparation + construction

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
Materials Usage	B.6	Maximize the re-use of earth cut materials, excess spoils, Construction debris / waste from within work sites; ensure usage results in no interference to natural drainage pattern	new S/S + existing S/S	Materials sources / stocks	Ocular inspection / spot checks / Materials log-book	Weekly
Materials / Stockpiling	B.7	Stockpiles to be placed within demarcated areas within the work sites that are fenced / barricaded; dust prone materials to be stored in enclosures and/or covered with tarp	new S/S + existing S/S	Materials sources / stockpile storage area	Ocular inspection / spot checks / Materials log-book	Weekly
	B.8	Implement S-EMP: Materials Management Plan (including warehouses / storage)	new S/S + existing S/S	S-EMP implementation	Materials Log-book	Weekly
Borrow areas / spoils disposal areas	B.9	Use licensed / authorized borrow areas / spoils disposal areas	area of influence	Use of licensed / authorized areas / sites	Copy of license / authorization paper on file; records maintained	Weekly during construction or as required during project activity
	B.10	Disposal of spoils (if any) on paddy fields, farmland, or within 50 m of a water source is prohibited	area of influence	Farmland or water source in the wider area of influence	Ocular inspection / spot checks / log-book with number of trips and spoil disposal location	Weekly during construction or as required during project activity
	B.11	Ensure vehicles are covered during transportation of stockpiles, spoils, etc., to avoid spillage	area of influence	Provision for vehicles cover	Ocular inspection	During transportation
	B.12	Implement S-EMP: Spoils Disposal	new S/S	S-EMP implementation	Log-book with number of trips and spoil disposal location	Weekly during construction or as required during project activity

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
	B.13	Rehabilitate / vegetate spent borrow areas / spoils disposal areas within 30 days of closure (to prevent soil erosion and dust)	borrow pits / spoils sites	Rehabilitated areas	Ocular inspection / spot-checks	Within 30 days of closure
Soil erosion	B.14	Restore loose soil from foundation work at work sites through ramming if required; avoid erosion by levelling site after land and vegetation clearance	new S/S + existing S/S	Erosion control	Ocular inspection	Weekly
Presence of construction (workers) camps	B.15	Construction (workers) camps maintained in demarcated areas as per requirements and measures A.3, A.4 and A.5, Table 2 and in accordance with IFC (WBG) Guidance Note / Workers Accommodation	new S/S	Construction (workers) camps	Ocular inspection / spot checks	Weekly
Land levelling	B.16	Ensure proper land levelling, closing of any excavations and/or trenches, grading for soil stabilization, drainage and other erosion prone areas including permanent stabilization measures at least within 30 days of completion of civil works	new S/S + existing S/S	Habitat restoration, erosion control	Ocular inspection	Once within 30 days of completion of construction
Landscaping	B.17	Implement landscaping along the fenced perimeter of new S/S, maintain hedges and field margins (thereby reducing visual and dust impact)	new S/S	Landscaped areas within new S/S	Ocular inspection	Once after completion of construction
	B.18	Implement S-EMP: Site Rehabilitation and Clean-up plan to improve visual aesthetics	new S/S	S-EMP implementation	Log-book	Once after completion of construction

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
Damage to infrastructure	B.19	Any damage to areas or infrastructure outside the agreed work sites (corridor of impact assessed as per social safeguards due diligence) to be restored to pre-construction conditions and will be subject to compensation at EPC contractor cost and through written agreement with the land owner / relevant agency, as applicable ²⁵	new S/S + existing S/S	Part of the contract clause	Ocular inspection	Once after completion of construction
		Stormwater, Drainage Management				
Disturbance to natural drainage pattern due to civil works	C.1	Backfilling, use / re-use of materials or excess spoils, placement of stockpiles to take due cognizance of, maintain the natural drainage pattern and causes no disturbance to nearby existing natural water sources / bodies, drainage, bunds, etc; Keep loose soil material and stockpiles out of drains and flow-lines)	new S/S	Natural drainage pattern maintained; drains and flow-lines cleaned and maintained	Ocular inspection / spot checks	Weekly during construction + once after completion of construction
		Water Resources, Water Quality				
Impact on water resources / groundwater sustainability	D.1	Water usage for civil works / construction as per stipulated quantity or volume sanctioned by the local water supply service provider	new S/S	Memorandum of Understanding (MOU) between contractor and service provider	Project specific commitment letter on file	Once

²⁵ Measures in B.17, Table 3 to be undertaken in coordination with the PIU

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
Run-off from general construction activities, spills, stockpiles; presence of construction (workers) camps	D.2	All work sites, construction (workers) camps, materials handling and storage areas, waste collection area, stockpiles, etc shall be at least 10 m distance from water sources/ bodies, drainage, bunds; Keep loose soil material and stockpiles out of drains and flow-lines	new S/S	10 m distance maintained; drains and flow-lines cleaned and maintained	Ocular inspection / spot checks	Weekly
	D.3	Provision of temporary adequate receiving body (e.g., bunds, tanks) at work sites without causing onsite / offsite adverse impacts from construction run-off and/or water logging; After collection, process water and muddy runoff and if necessary, employ flocculants such as PAM (Polyacrylamide) to facilitate sedimentation or filter to remove silts	new S/S + existing S/S	Provision of an adequate receiving body (bunds, tanks) for construction run-off onsite	Ocular inspection	Once
	D.4	Provision and maintenance for temp sanitation (portable toilets) facilities for workers (separate for genders) in construction (workers) camps	construction (workers) camps	Provision of temp sanitary facilities	Ocular inspection / spot-checks / Maintenance log-book	Daily
	D.5	Collection of sanitation waste by licensed vendor, to be transported by specialized trucks with units to nearest waste disposal facilities	construction (workers) camps	Memorandum of Understanding (MOU) between contractor and licensed vendor	Project specific commitment letter on file; record of number of trips maintained	Weekly or as required during project activity
	D.6	Conduct surface water quality monitoring	any natural receiving water source/ body that is immediately adjacent to new S/S and/or	Surface water quality monitoring conducted	Sampling and chemical analysis	Quarterly during construction + once after completion of construction

Table 3. Construction Stage: New Substation (S/S) + System Strengthening and Augmentation Works of Existing Substations (S/S)						
Key Activities / Potential Impacts & Issues during Construction Stage	No.	Mitigation Measures and Safeguards	Monitoring Location ²⁴	Monitoring Parameter	Method of Measurement	Frequency
			downstream of new S/S			
	D.7	Comply with relevant National standards on water quality / IFC (WBG) Environmental, Health, and Safety (EHS) Guidelines on Wastewater and Ambient Water Quality, Electric Transmission and Distribution ²⁶	new S/S	Compliance with National standards or IFC (WBG) Guidelines, whichever is stringent	Record and maintain log of monitoring / incidences of non-compliance and rectification	Quarterly during construction + once after completion of construction
	D.8	Implement S-EMP: Construction Wastewater	new S/S + existing S/S	S-EMP implementation	Log-book	Daily
		Air Quality / Fugitive Emissions				
Increase in air emissions (including fugitive dust) due to movement of heavy equipment, machinery, construction vehicles, civil works (levelling, grading, back-filling, concrete / cement mixing, etc), stockpiles, etc.	E.1	Periodic watering (water spraying) before movement of vehicles, at work sites, open and exposed areas and access roads, exposed stockpiles (if any), during concrete / cement mixing process	new S/S + existing S/S	Spraying of water	Ocular inspection / spot checks	Daily during construction in dry season

²⁶ <https://www.ifc.org/en/insights-reports/2000/general-environmental-health-and-safety-guidelines>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-general-ehs-guidelines-wastewater-and-ambient-water-quality-en.pdf>; <https://www.ifc.org/content/dam/ifc/doc/2000/2007-electric-transmission-distribution-ehs-guidelines-en.pdf>